

Luke Zheng

lukez8563@gmail.com • (858)-375-8496 • LinkedIn.com/in/luke-zheng1

Education

University of California, San Diego

Expected: June 2026

Bachelor of Science in **Computer Science**, Minor in Business-Economics

GPA: 3.7/4.0

- **Relevant Coursework:** Software Engineering, Computer Operating Systems, Data Structures and Object-Oriented Design, Computer Security, Recommender Systems & Web Mining, Data Science, ML Learning Algorithms

Experience

Vivo | Machine Learning Engineer Intern

San Diego, CA | Sep 2024 – Present

- Transformed raw video, depth, and trajectory data into 18.2MB 3D point clouds (~9K points each) using open-source tools, correcting misalignments and validating outputs to deliver reliable 3D data for AI research
- Tested 3 alternative 3D representations through 10+ model runs and 5 detailed metric comparisons to assess data quality, supporting 4 collaborative team discussions on improving inputs for more accurate scene reconstruction
- Coordinated 10+ cross-functional sessions with researchers and engineers to document 15+ workflow updates, standardize processes, and implement guidelines, ensuring reproducible pipelines for AI research experiments

Vivo | Software Engineer Intern

San Diego, CA | Jun 2024 – Sep 2024

- Processed 100+ video inputs to produce codebook features used in MAMBA-based models, delivering structured inputs that improved training stability and enabled longer, higher-quality AI-generated videos
- Improved data quality for scene modeling by debugging errors in the extraction pipeline, rewriting ~50 lines of code, and testing 100+ video samples to ensure accurate, recognizable 3D representations for research use
- Enabled reliable model evaluation by developing 3 benchmarking scripts using Fréchet Video Distance, processing 200+ video samples to deliver consistent performance comparisons for AI-generated video research

UCSD Graduate/Undergraduate Economics Lab | Research Assistant

La Jolla, CA | Oct 2023 – Jun 2024

- Developed a Chrome extension to enable automated entry of 18 input fields across 15 major job application platforms, cutting form completion time by ~50% for 18+ users and boosting overall productivity through streamlined workflows
- Led 20+ weekly agile sprint reviews with research mentors to prioritize 10+ feature enhancements, refine usability for 18+ target users, and maintain clear documentation, improving alignment with real-world job application needs
- Built a deployment-ready automation tool with 30+ Regex, dynamic mapping for 18 input fields, and a 3-page user manual, simplifying adoption, supporting non-technical users, and enabling consistent use across 15+ job platforms

Projects

Goodreads Recommender System

- Built a Python recommendation system predicting user-book interactions (~76.5% accuracy) with Jaccard similarity and popularity features; implemented **logistic regression** using scikit-learn, NumPy, and pandas on 200K+ reviews
- Developed bias-based collaborative filtering to predict star ratings, tuning regularization and learning rates to reduce error; used **pandas**, **NumPy**, and **scikit-learn** for preprocessing, training, and evaluation

CrimeFinder

- Engineered a scalable backend system by ingesting 800,000+ crime incident records into a PostgreSQL database and implementing a RESTful API to enable real-time data access, filtering, and query handling
- Integrated OpenAI GPT-3.5 via web API to classify crime severity with natural language outputs; built a frontend visualization layer using Leaflet.js, optimizing map rendering to sub-5s latency on 100K+ data points

Leadership & Activities

Triton Business Society | Senior Vice President

La Jolla, CA | Oct 2024 – Jun 2025

- Managed internal operations for a 50+ member organization by updating bylaws, planning 20+ meeting agendas, overseeing elections, and coordinating 5+ budget reviews, ensuring smooth governance and consistent execution
- Addressed attendance challenges by tracking weekly participation metrics, led 10+ targeted marketing initiatives, and built 5 partnerships through coordinated outreach, boosting event participation by 30% and enhancing campus visibility

Product Management Club | Product Advisor

La Jolla, CA | Mar 2024 – Jun 2025

- Managed agile workflows during a 9-week client project coordinating 5 content marketers, analysts, and designers, tracking weekly deliverables, and facilitating 6 review sessions to deliver branding and strategy recommendations
- Conducted detailed market research on 3+ key competitors (e.g., Freed.ai), shaped product positioning and outreach strategy, and led 4 iterative design reviews to produce tailored marketing assets that boosted stakeholder alignment

Skills & Interests

Languages & Tools: Python, Java, JavaScript, C++, SQL, HTML/CSS, React, Flask, Node.js, Leaflet.js, MaterialUI, PyTorch, TensorFlow, NumPy, scikit-learn, Git, GitHub, VSCode, Jupyter Notebook, MatLab, Agile, JSON, JUnit

Interests: Racket sports, rock climbing, National Parks, collectible market analysis, computer vision, AI applications